1. These components of the computer hardware actively and concurrently access the shared memory through a shared bus.
   Direct Memory Access Device
   Central Processing Unit

2. The operating system is needed because _______.
   I/O operations are frequent and slow
   CPU cannot be kept busy all the time by ANY application

3. Check all statements that apply to the DMA.
   DMA is a hardware device
   DMA is faster than the CPU to perform data transfers

4. These components of the computer hardware actively and simultaneously access the shared memory through a shared bus.
   None of the above

5. True of False? On modern general purpose computer systems, software may directly trigger and interrupt onto the CPU.
   False

6. On a computer hardware with multiple CPUs and a shared memory and bus system, I/O devices and the CPUs execute ______.
   concurrently

7. Software may trigger an interrupt by executing a special operation called a ______.
   system call

8. ___________ may trigger an interrupt at any time by sending a signal to the __________.
   hardware
   CPU

9. The ________ loads at start up the ________.
   bootstrap program
   operating system kernel

10. The ______ is small and loaded at start up or reboot. It is NOT initially stored in the _______.
    bootstrap program
    RAM

11. The CPU _______ the data transfers from/to main memory to/from local buffers.
    initiates

12. The bootstrap program is precisely ________.
    a piece of firmware